

COIMBATORE INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION AFFILIATED TO ANNA UNNIVERSITY)



Online Chit Fund Management

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Online Chit Fund Management

Project Objective

The objective of the project is to develop an online platform for efficient management of chit funds. This platform aims to streamline the processes involved in chit-fund operations, providing convenience and transparency to both chit-fund organizers and participants.

Project Scope

The scope of the project includes designing and implementing a user-friendly web application that facilitates various functionalities such as chit fund creation, participant enrollment, bidding process, payment management, documentation, and reporting.

It facilitates easy online chit fund from home for both customer and admin side

It avoids all manual maintenance of data

Problem Statement

Traditional chit fund management systems are often cumbersome, involving manual paperwork and inefficient communication channels. There is a need for a modernized approach to manage chit funds, leveraging technology to enhance accessibility, transparency, and efficiency.

Problem Definition

The problem lies in the lack of centralized, automated systems for chit fund management, leading to inefficiencies, delays, and potential discrepancies in the process. This necessitates the development of an online platform to address these challenges and streamline operations.

Project Description

The project involves the development of a comprehensive online chit fund management system, utilizing a range of functional tools and adhering to key

Program: ChitFundAuctionSystem

Functional Keys

Database Connection: Establish connection to MySQL database using JDBC driver. Execute SQL queries to insert dividend details.

User Interface: Create a JFrame with title, size, and default close operation. Setup JPanel with components for user input and display of results. Handle button click to calculate dividend.

Calculation Logic: Calculate dividend based on discount percentage and pot value. Save dividend details to database.

Non-Functional Tools

Error Handling: Catch and handle exceptions during database connection and dividend calculation. Display error messages using JOptionPane.

Data Validation: Validate discount percentage input to ensure it falls within the allowed range.

Program: CustomerLogin

Functional Keys

Database Connection: Connect to MySQL database using JDBC driver. Execute SQL queries to retrieve customer password for authentication.

User Interface: Create a JFrame for customer login with text fields and password field. Handle button click to authenticate customer.

Authentication Logic: Verify customer credentials by comparing input password with stored password in database.

Non-Functional Tools

Error Handling: Catch and handle exceptions during database connection and login authentication. Display error messages using JOptionPane.

Data Validation: Validate customer ID format and password presence.

Program: CustomerRegistration

Functional Keys

Database Connection: Connect to MySQL database using JDBC driver. Execute SQL queries to insert new customer details.

User Interface: Create a JFrame for customer registration with text fields and password field. Handle button click to register new customer.

Registration Logic: Insert new customer details into database.

Non-Functional Tools

Error Handling: Catch and handle exceptions during database connection and customer registration. Display error messages using JOptionPane.

Data Validation: Validate input fields to ensure correct format (e.g., email, phone number).

Program: JoinChit

Functional Keys

Database Connection: Connect to MySQL database using JDBC driver. Execute SQL queries to load chit groups and insert customer into a chit group.

User Interface: Create a JFrame for joining chit groups with a table displaying available groups. Handle table row selection to show confirmation dialog for joining a group.

Joining Logic: Check if the customer is already a member of the selected chit group before joining. Update available slots in the chit group after joining.

Non-Functional Tools

Error Handling: Catch and handle exceptions during database connection and chit group joining. Display error messages using JOptionPane.

Program: ManageAuctions

Functional Tools Used

Swing Components: Utilized Swing components (JFrame, JPanel, JLabel, JButton) for creating the graphical user interface (GUI).

ActionListener: Implemented ActionListener interfaces to handle button click events for conducting new auctions and viewing existing auctions.

JOptionPane: Utilized JOptionPane to display a message dialog for informing the user about viewing existing auctions.

Layout Management: Used GridLayout to organize the components in a grid layout for better visual presentation.

Non-Functional Aspects

User Experience: Provides a user-friendly interface for managing auctions with clearly labeled buttons.

Resource Management: Properly disposes of the JFrame instances after use (DISPOSE_ON_CLOSE) to manage system resources efficiently.

Responsive Design: Adapts to different screen sizes with a fixed-size layout (setSize (400, 300)) for consistent display across devices.

Error Handling: Although no explicit error handling is implemented, the application ensures a smooth user experience by not crashing in case of errors.

Program: ManageChitGroups

Functional Tools Used

Swing Components: Utilized Swing components (JFrame, JPanel, JLabel, JButton, JTable, JScrollPane, GridLayout) for creating the graphical user interface (GUI).

ActionListener: Implemented ActionListener interfaces to handle button click events for adding new chit groups and selecting existing chit groups.

JOptionPane: Utilized JOptionPane to display error messages and information dialogs for user interaction.

Database Connectivity: Established database connections using JDBC (Connection, PreparedStatement, ResultSet) for retrieving and storing chit group data in the MySQL database.

Random Number Generation: Employed Random class to generate random alphanumeric group IDs for newly added chit groups.

Non-Functional Aspects

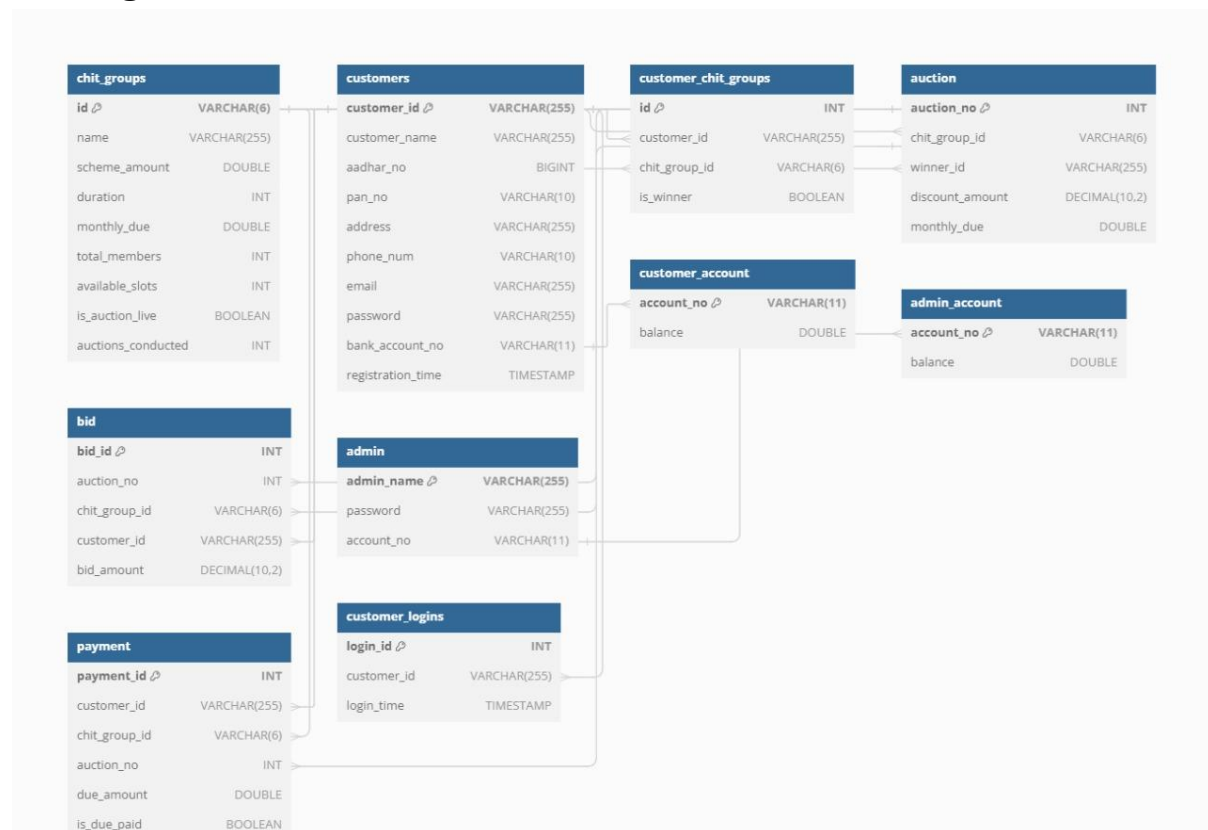
User Experience: Provides a user-friendly interface for managing chit groups with clear buttons and table layout.

Error Handling: Implements error handling for empty fields and invalid numerical formats, ensuring smooth user interaction.

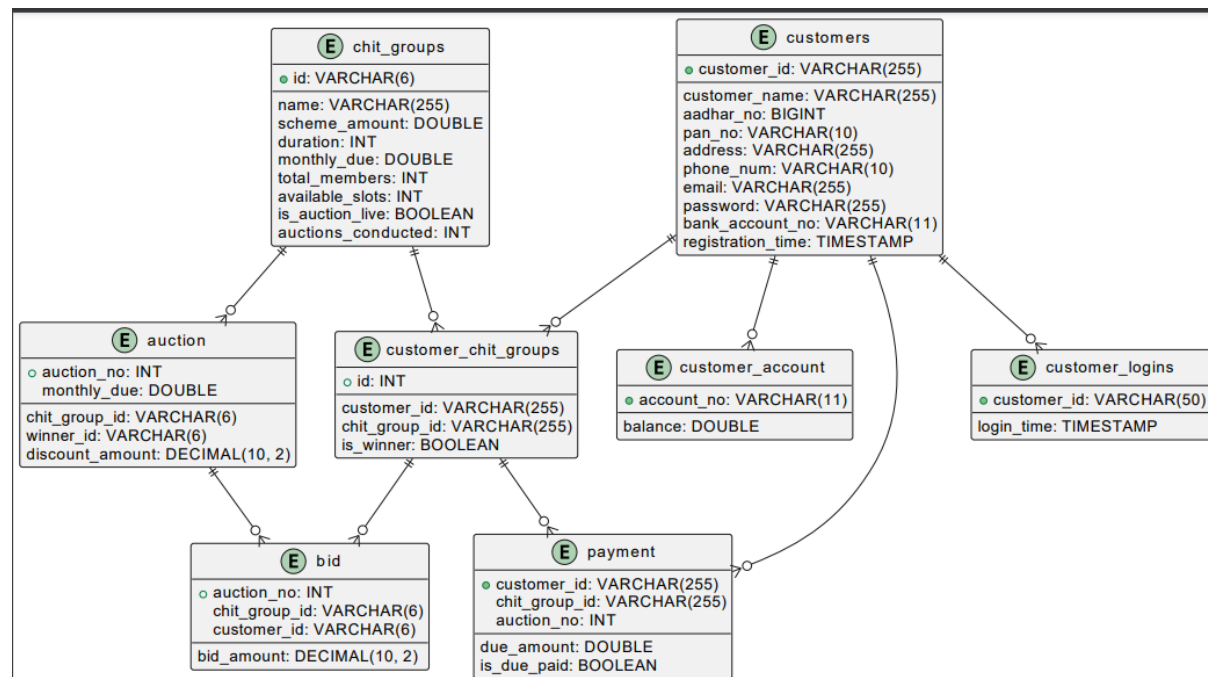
Resource Management: Properly disposes of JFrame instances after use (DISPOSE_ON_CLOSE) to manage system resources efficiently.

Responsive Design: Adapts to different screen sizes with fixed-size layouts (setSize) and centered dialog placement (setLocationRelativeTo(null)) for consistent display across devices.

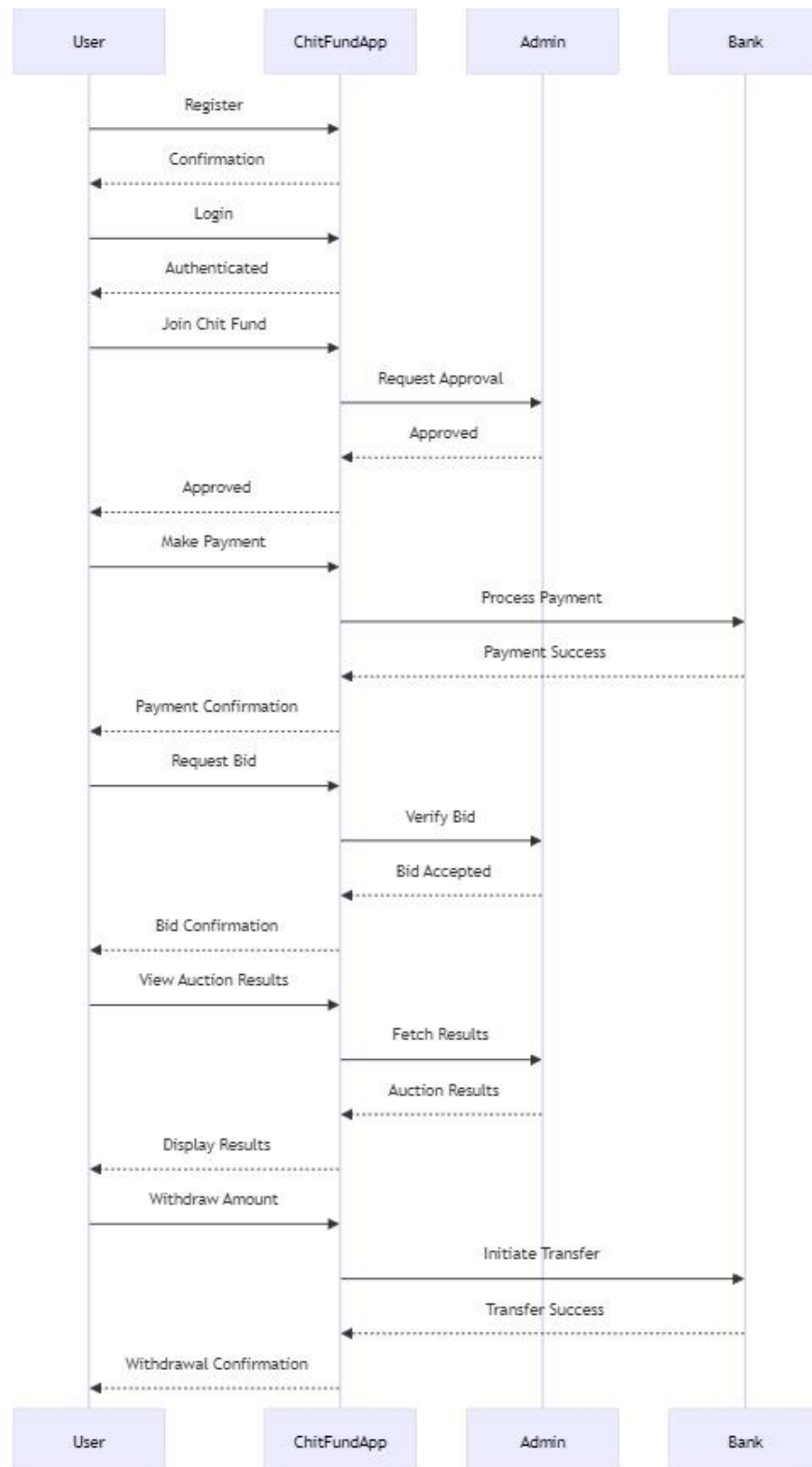
ER diagram



Class diagram:



Sequence diagram:



Admin login:

A screenshot of a web browser window titled "LOGIN". The background is a collage of US dollar bills. In the center, there is a white spiral-bound notepad with a grid pattern. On the notepad, the text "ADMIN LOGIN" is displayed in bold. Below it, there are two input fields: "ADMIN ID" and "PASSWORD". At the bottom of the notepad, there is a dark blue button with the text "LOGIN" in white.

ADMIN LOGIN

ADMIN ID

PASSWORD

LOGIN

Admin Panel:

A screenshot of a web browser window titled "Admin Panel". The background is a collage of US dollar bills. In the center, there are three white buttons with black text, stacked vertically. The buttons are labeled "MANAGE CHIT GROUPS", "MANAGE CUSTOMERS", and "MANAGE AUCTIONS".

MANAGE CHIT GROUPS

MANAGE CUSTOMERS

MANAGE AUCTIONS

Manage Chit Groups:

Manage Chit Groups

MANAGE CHIT GROUPS					
GROUP ID	NAME	SCHEME AMOUNT	DURATION	MONTHLY DUE	AVAILABLE SLOTS
QNUBA3	School Fee	50000.0	10	5000.0	0
91V0WP	Bike	100000.0	10	10000.0	0
P7QKV7	Gold	200000.0	20	10000.0	0
UJBCHU	Car	300000.0	30	10000.0	0



ADD CHIT GROUP

Create new chit groups:

Add Chit Group

Name :

Scheme Amount:

Tenure (months):

Monthly Due :

Total Members:

Add

Manage Customers:

Manage Customers

Customer Details							
Customer ID	Customer Name	Aadhar No	PAN No	Address	Phone Number	Email	Bank Accoun...
2D5GXL	Mughil	396742120909	OHDED1290W	1/24, EUUY STREET, IDO...	6238471239	arun@gmail.com	10102023331
2G1PFU	Jeeva	967371231122	ABCD89676F	191/43, SHDGFHD STREET...	9283723219	jeeva@gmail.com	10101010101
5MLC5B	Vijay	967698769876	ANMER1234I	5/34, NCEJF, UICFCFN + 9...	7654321098	vijay@gmail.com	99991111000
6QU5IU	Kaavya	19234341909	BDKXZ8292W	1/111, IURBD STREET, JDI...	9999112928	kaavya@gmail.com	10203404569
ANY13C	Vishal	121290901234	ASDFQ1010E	8/23, UEYDN STREET, IUN...	9289290101	vishal@gmail.com	11112222333
C4EQU	Shalini	232018381239	HGWJO9823O	9/120, IHUBP STREET, MJ...	7718347281	shalini@gmail.com	49134819707
DYSZQZ	Karthi	289718391721	ODHJY1781I	1/29, FKJGU STREET, OI...	9283763109	karthi@gmail.com	10107272110
KVNGU	Virat	138123722345	AKHDJ2131Q	2/13, KUGDEV STREET, UI...	9873323123	virat@gmail.com	87987098909
L9TFGS	John	463641203193	OHGHO1234O	2/199, PLPLP STREET, HO...	7878231210	john@gmail.com	20392997009
MBSXY	Ashok	111100002323	CPLP71717	4/444, LKHGD STREET, ...	9283221200	ashok@gmail.com	10120203030
OYG5OM	Sruthi	553627181922	JHGDH1234M	3/319, UHYBVSO STREET...	6298328973	sruthi@gmail.com	53782163909
QOQGGD	Priya	387482110230	LKOIP1943D	2/222, EBBBBX STREET, ...	8178781839	priya@gmail.com	73647492123
S7LWWZ	Ajith	123412341234	ABCDEF1234	1/234, QWERTY, ASDFGH ...	9087654322	ajith@gmail.com	98671111222
VTQH48	Raji	238190202819	HBKQU1299S	6/43, IHBQB STREET, LINI...	9091091091	raji@gmail.com	90128910107
YK8SIR	Surya	456745674567	ABTED1234L	9/43, ZCXXVV STREET, E...	8768751234	fzzztrs@gmail.com	12345678901

Manage auction:

New Auction

Bike

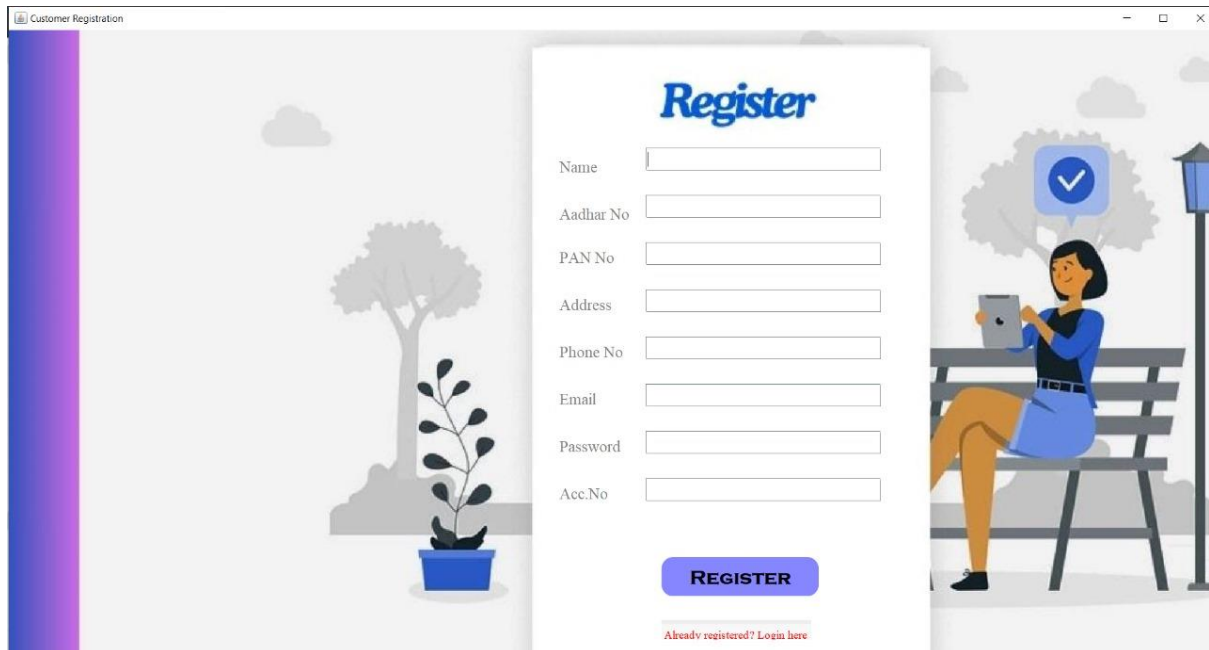
CONDUCT AUCTION

END AUCTION

VIEW AUCTION

CUSTOMER SIDE APPLICATION:

Customer registration:



A screenshot of a web browser window titled "Customer Registration". The page features a light gray background with a stylized illustration of a person sitting on a bench and using a tablet. The registration form is centered and includes the following fields: Name, Aadhar No, PAN No, Address, Phone No, Email, Password, and Acc.No. A blue "REGISTER" button is positioned below the form. A link "Already registered? Login here" is located at the bottom of the form. The form is set against a background that includes a potted plant and a tree.

Register

Name

Aadhar No

PAN No

Address

Phone No

Email

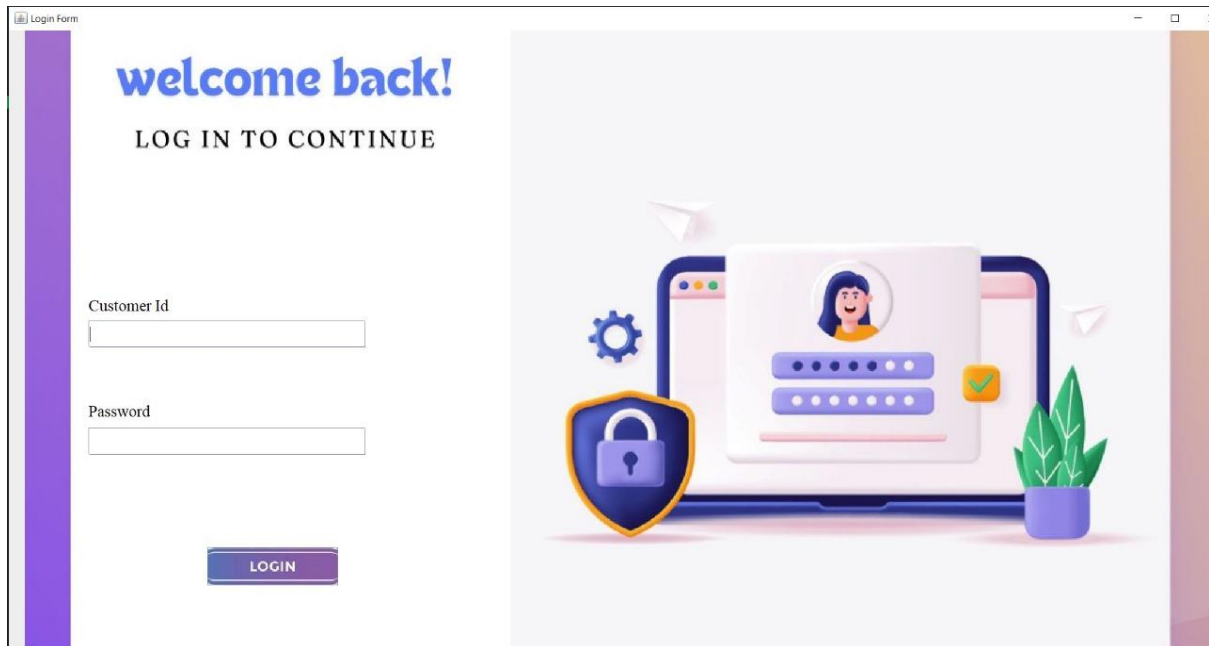
Password

Acc.No

REGISTER

[Already registered? Login here](#)

Customer login:



A screenshot of a web browser window titled "Login Form". The page features a light gray background with a stylized illustration of a laptop displaying a user profile, a shield with a lock, and a potted plant. The login form is on the left and includes the following elements: a "welcome back!" message, a "LOG IN TO CONTINUE" instruction, a "Customer Id" field, a "Password" field, and a blue "LOGIN" button.

welcome back!

LOG IN TO CONTINUE

Customer Id

Password

LOGIN

View Available chit:

Join Chit Group

EXISTING CHIT GROUPS

GROUP ID	NAME	SCHEME AMOUNT	DURATION	MONTHLY DUE	TOTAL MEMBERS	AVAILABLE SLOTS
						
VIEW MY CHIT GROUPS						

View our chit groups:

My Chit Groups

MY CHIT GROUPS

GROUP ID	NAME	SCHEME AMOUNT	TENURE	MONTHLY DUE	TOTAL MEMBERS	AVAILABLE SLOTS	AUCTIONS CON...
91V0WP	Bike	100000.0	10	10000.0	10	0	0
P7QKV7	Gold	200000.0	20	10000.0	10	0	0
QNUBA3	School Fee	50000.0	10	5000.0	10	0	0



JOIN AUCTION

Future Enhancements:

In the future, online chit fund management systems could incorporate features like AI-powered risk assessment for better decision-making, blockchain technology for enhanced security and transparency, automated payment reminders, and advanced data analytics for predicting member behavior and optimizing fund operations. Additionally, integrating mobile apps for easy access and real-time updates could further enhance user experience.

We can integrate payment gateways like razorpay. Swapping of chit groups in case of vacancy

Conclusion

The development of an online chit fund management system addresses the inefficiencies and challenges inherent in traditional approaches. By leveraging technology, this platform enhances accessibility, transparency, and efficiency in chit fund operations, benefiting both organizers and participants. Moving forward, continuous refinement and adaptation will be essential to meet evolving user needs and market dynamics.